Data Science with Python 3

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Chapter One -Introduction to Data Science

In this course you will learn dealing, manipulating data, visualize it, make decisions.

You'll learn to take data:

- **Process it**
- **Visualize it**
- **Understand it**
- **Communicate it**
- **Extract value from it**

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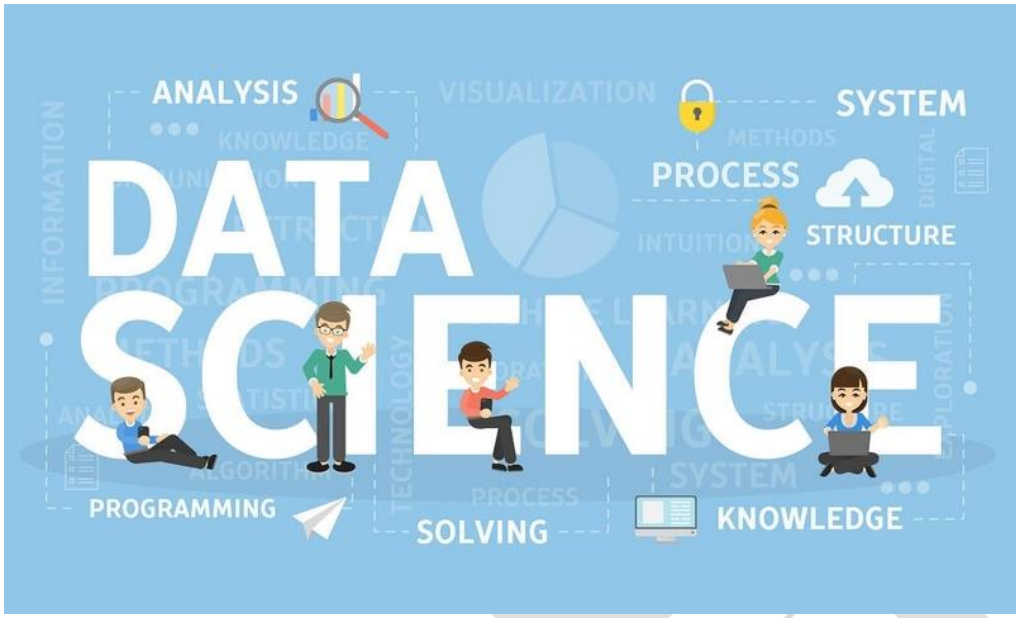
يتطور **علم البيانات** بشكل متسارع ليصبح واحداً من أهم المجالات في صناعة التكنولوجيا. يرجع الفضل للتقدم السريع في مجالات عدة كالاتصالات و الخدمات السحابية وكفاءة أجهزة الحاسب والذي جعل المستحيل ممكناً. **من الممكن الآن تحليل مجموعات كبيرة من البيانات الضخمة والتي بدورها تمكننا و إلى حد غير مسبوق من اكتشاف حقائق و أنماط ورؤى ربما يستحيل اكتشافها بدون تحليل البيانات**. مثال على هذه البيانات، سجلات زوار موقع إلكتروني بيانات المرضى.

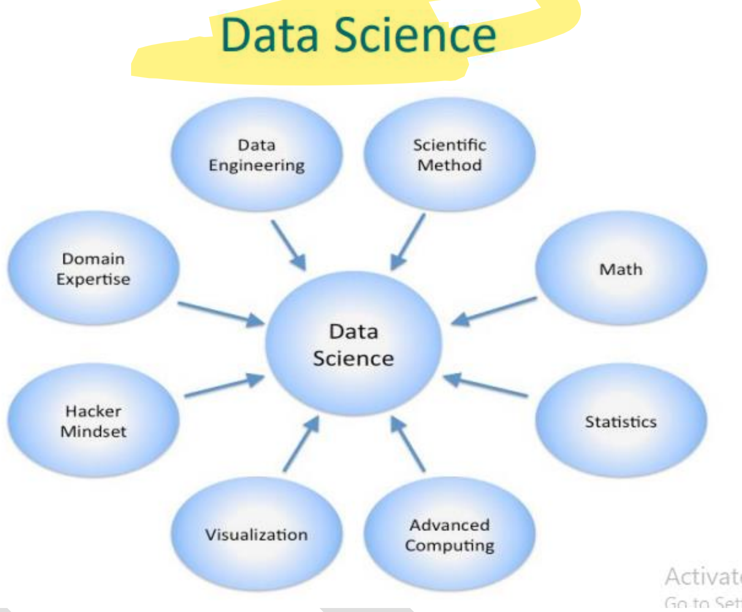
- محلل البيانات
- علماء البيانات
- مهندس البيانات
- مهندس تعلم الالة



Essential Python Libraries: **NumPy, pandas, matplotlib, SciPy, scikit-learn**

Science Fields related to Data Science:

Statistics + **mathematics** + **computer science** + **operations research** + **statistical and machine learning** + **data warehousing** + **data visualization** + **information science** + **Big data**, …



Before starting to learn Data Science, you need to learn some *terminologies*. You need to know **What is Data Science**?

*"The ability to take **data**—to be able to **understand** it, to **process** it, to **extract** value from it, to **visualize** it, to **communicate** it—that's going to be a hugely important skill in the next decades, not only at the professional level but even at the educational level for elementary school kids, for high school kids, for college kids."*

Data Science is

- An area that manages, manipulates, extracts, and interprets knowledge from tremendous amount of data
- Data science (DS) is a multidisciplinary field of study with goal to address the challenges in big data
- Data science principles apply to all data – big and small

Theories and techniques from many fields and disciplines are used to investigate and analyze a large amount of data to help decision makers in many industries such as science, engineering, economics, politics, finance, and education.

– Computer Science

• Pattern recognition, visualization, data warehousing, High performance computing, Databases, AI

– Mathematics

• Mathematical Modeling

– Statistics

• Statistical and Stochastic modeling, Probability.

Data All Around:

Lots of data is being collected and warehoused

– Scientific Experiments

– Internet of Things

– Web data, e-commerce

– Financial transactions, bank/credit transactions

– Online trading and purchasing

– Social Network

– ……many more!

We live in a world that's drowning in data.

• Sensing devices and sensor networks that can monitor everything 24/7 from temperature to pollution to vital signs

• Increasingly sophisticated smart phones

• Internet, social networks makes it easy to publish data

• Scientific experiments and simulations à astronomical data volumes

• Internet of Things

البيانات تحيط بنا:

يتم جمع الكثير من البيانات وتخزينها

- بيانات الويب ، التجارة الإكترونية
- المعاملات المالية ، والمعاملات المصرفية / الإئتمان
- التداول والشراء عبر الإنترنت
- الشبكات الإجتماعية


How to handle that data? How to extract interesting actionable insights and scientific knowledge?

Websites track every user's every click. Your smartphone is building up a record of your location and speed every second of every day. "Quantified selfers" wear pedometers-on-steroids that are always recording their heart rates, movement habits, diet, and sleep patterns.

Smart cars collect driving habits, smart homes collect living habits, and smart marketers collect purchasing habits. The internet itself represents a huge graph of knowledge that contains (among other things) an enormous cross-referenced encyclopedia; domain-specific databases about movies, music, sports results, pinball machines, memes, and cocktails; and too many government statistics (some of them nearly true!) from too many governments to wrap your head around.

- تتعلم الشركات أسرارك وأنماط التسوق والتفضيلات الخاصة بك: على سبيل المثال ، هل يمكننا معرفة ما إذا كانت المرأة حامل ، من أنماط تسوقها
- الإعلانات المدفوعة على FaceBook تستهدف مجموعات معينة .
- التصفح علىGoogle

Data Science: Case Studies:

1- Cancer Research

Cancer is an incredibly complex disease; a single tumor can have more than **100** billion cells, and each cell can acquire mutations individually.
The disease is always changing, evolving, and adapting.

- Employ the power of big data analytics and high-performance computing.
- Leverage sophisticated pattern and machine learning algorithms to identify patterns that are potentially linked to cancer
- Huge amount of data processing and recognition

2- Health care

Healthcare is an incredibly complex issue; ……

3- Elections

In 2012, the Obama campaign employed dozens of data scientists who datamined and experimented their way to identifying voters who needed extra attention, choosing optimal donor-specific fundraising appeals and programs, and focusing get-out-the-vote efforts where they were most likely to be useful. And in 2016 the Trump campaign tested a staggering variety of online ads and analyzed the data to find what worked and what didn't.

قام مليون شخص بتثبيت تطبيق أوباما على Facebook والذي أتاح الوصول إلى المعلومات عن "الأصدقاء"

4- Internet of Things (IoT)

The Internet of Things is rapidly growing. It is predicted that more than 40 billion devices will be connected by 2025.

5- Customer Analytics

Customer Analytics is ….

```
users = [
    { "id": 0, "name": "Hero" },
    { "id": 1, "name": "Dunn" },
    { "id": 2, "name": "Sue" },
    { "id": 3, "name": "Chi" },
    { "id": 4, "name": "Thor" },
    { "id": 5, "name": "Clive" },
    { "id": 6, "name": "Hicks" },
    { "id": 7, "name": "Devin" },
    { "id": 8, "name": "Kate" },
    { "id": 9, "name": "Klein" } ]

    friendship_pairs = [(0, 1), (0, 2), (1, 2), (1, 3), (2, 3),
                        (3, 4), (4, 5), (5, 6), (5, 7), (6, 8), (7, 8), (8, 9)]
```

Data life cycle:

- **Data collecting**

تحديد مصادر البيانات المختلفة، والتي قد تشمل سجلات **خادم الويب** ، أو منشورات **الوسائط الاجتماعية،** أو البيانات من المكتبات الرقمية مثل مجموعات بيانات التعداد السكاني في الولايات المتحدة، أو البيانات التي يتم الوصول إليها من خلال المصادر الموجودة على الإنترنت عبر واجهات برمجة التطبيقات، أو المعلومات الموجودة بالفعل في جدول بيانات.(Excel)

- **Data Preparing/Processing**

جمع بيانات مثل اختيار البيانات ذات الصلة، ودمجها عن طريق خلط مجموعات البيانات، وتنظيفها، والتعامل مع القيم المفقودة إما عن طريق إزالتها أو احتسابها مع البيانات ذات الصلة، والتعامل مع البيانات غير الصحيحة عن طريق إزالتها، وكذلك التحقق من القيم المتطرفة والتعامل معها.

- **Data Analysis**
- **Analysis hypothesis**
- **Decision**

1.0 OBJECTIVES

1. Introduction to Data Science

2. Familiarize with the issues and challenges in Data Growth

3. Familiarize with data Science Process

4. Familiarize with the concepts of data like **clean** and **messy** data, **data artifacts** and anomalies in data.

1.1 DATA SCIENCE-A DISCIPLINE

Data Science is a blend of various tools, algorithms and machine learning principles with the goal to discover hidden patterns from raw data. It is related to data mining, machine learning and Big Data.

It is an area that manages, manipulates, extracts, and interprets knowledge from tremendous amount of data.

Data science (DS) is a multidisciplinary field of study with a goal to address the challenges in big data. Big Data has given rise to Data Science and is a therefore, a multidisciplinary field of study with goal to address the challenges in big data.

علم البيانات هو مجال دراسي **متعدد التخصصات** يهدف إلى معالجة التحديات في البيانات الضخمة. وقد أدت البيانات الضخمة إلى ظهور علم البيانات، وبالتالي فهو مجال يهدف إلى معالجة التحديات في البيانات الضخمة.

1.2 DATA TO DATA SCIENCE

The growth of data has been seen since 2010, due to the growth in the number of data generating devices like smart phones, wearables, Internet of things, etc. The availability of more data publicly from **social media sites** like **Facebook**, **YouTube**, **twitter** etc, business transactions, sensors, audio, video, photos etc. This enormous growth of data led to the concept of Big data, which is a term used for collection of large and complex data sets.

As the **data** has **increased**, so did the need for its **storage**. Until 2010, the main focus was building framework and solutions to store data, which was successfully solved by **HADOOP** and other frameworks. In the present time, it has become difficult to process this large and complex data using traditional data management techniques such as, for example, the RDBMS (relational database management systems).

This rise in the use of data, sparked حرض the use of Data Science. Data science makes this possible, as it is a multidisciplinary study of data collection for analysis, prediction, learning and prevention. Data Science involves using methods to analyze massive amounts of data and extract the **knowledge** from the **raw data**.

1.3 DATA GROWTH ISSUES AND CHALLENGES

Big Data is characterized by five **V's** namely **volume**, **variety**, **velocity**, **veracity** and **value**. Consequently, the challenges these characteristics bring are being seen in ***data capture***, ***curation***, ***storage***, ***search***, ***sharing***, ***transfer***, and ***visualization***.

تتميز البيانات الضخمة **بخمس** خصائص هي الحجم والتنوع والسرعة والمصداقية والقيمة. وبالتالي، فإن التحديات التي تفرضها هذه الخصائص تظهر في التقاط البيانات وتنظيمها وتخزينها والبحث عنها ومشاركتها ونقلها وتصورها.

i. **Volume** refers to the enormous size of data. Big data refers to data volumes in the range of exabytes and beyond e.g. In the year 2016, the estimated global mobile traffic was 6.2 Exabytes (6.2 billion GB) per month. Also, by the year 2020 we will have almost 40000 Exabytes of data. Such volumes exceed the capacity of current on-line storage systems and processing systems. Traditional database systems is not able to capture, store and analyse this large amount of data. Storage solutions have been provided by HADOOP framework, but represent long term challenges that require research and new paradigms.

يشير volume إلى الحجم الهائل للبيانات. تشير البيانات الضخمة إلى أحجام البيانات في نطاق **الإكسابايت** وما بعده على سبيل المثال في عام 2016، كان حجم حركة الهاتف المحمول العالمية المقدر 6.2 إكسابايت (6.2 مليار جيجابايت) شهريًا. أيضًا، بحلول عام 2020 سيكون لدينا ما يقرب من 40000 إكسابايت من البيانات. تتجاوز هذه الأحجام سعة أنظمة التخزين عبر الإنترنت وأنظمة المعالجة الحالية. لا تستطيع أنظمة قواعد البيانات التقليدية التقاط وتخزين وتحليل هذه الكمية الكبيرة من البيانات. تم توفير حلول التخزين من خلال إطار عمل HADOOP، ولكنها تمثل تحديات طويلة الأجل تتطلب البحث والنماذج الجديدة.

ii. **Velocity** refers to the high speed of accumulation of data. There is a massive and continuous flow of data from sources like machines, networks, social media, mobile phones etc. This determines the potential of data that how fast the data is generated and processed to meet the demands e.g. there are more than 3.5 billion searches per day are made on Google. Also, FaceBook users are increasing by 22%(Approx.) year by year. Data is streaming in at unprecedented speed and must be dealt with in a timely manner. RFID tags, sensors and smart metering are driving the need to deal with torrents of data in near-real time. The data has to be available at the right time to mske business decisions accurately. Reacting quickly enough to deal with data velocity is a challenge for most organizations.

تشير velocity إلى السرعة العالية لتراكم البيانات. هناك تدفق هائل ومستمر للبيانات من مصادر مثل الآلات والشبكات ووسائل التواصل الاجتماعي والهواتف المحمولة وما إلى ذلك. وهذا يحدد إمكانات البيانات ومدى سرعة إنشاء البيانات ومعالجتها لتلبية المتطلبات على سبيل المثال، هناك أكثر من 3.5 مليار عملية بحث يوميًا يتم إجراؤها على Google. كما يتزايد عدد مستخدمي Facebook بنسبة 22٪ (تقريبًا) عامًا بعد عام. تتدفق البيانات بسرعة غير مسبوقة ويجب التعامل معها في الوقت المناسب. تعمل علامات RFID وأجهزة الاستشعار والقياس الذكي على دفع الحاجة إلى التعامل مع سيول البيانات في الوقت الفعلي تقريبًا. يجب أن تكون البيانات متاحة في الوقت المناسب لاتخاذ قرارات العمل بدقة. يعد الاستجابة بسرعة كافية للتعامل مع سرعة البيانات تحديًا لمعظم المؤسسات.

iii. **Variety** refers to nature of data. Data is available in varied formats and heterogenous sources. It can be structured, semi-structured and unstructured data. It is a challenge to find ways of governing, merging and managing these diverse forms of data.

يشير التنوع variety إلى طبيعة البيانات. فالبيانات متاحة في أشكال متنوعة ومصادر **غير متجانسة**. ويمكن أن تكون بيانات مهيكلة وشبه مهيكلة وغير مهيكلة. ومن الصعب إيجاد طرق لإدارة ودمج هذه الأشكال المتنوعة من البيانات.

iv. **Veracity** here means quality of data. It refers to inconsistencies and uncertainty in data. The available data may be messy and thus controlling the quality and accuracy becomes another challenge.

الصدق Veracity هنا يعني جودة البيانات. ويشير ذلك إلى التناقضات وعدم اليقين في البيانات. فقد تكون البيانات المتاحة غير منظمة، وبالتالي يصبح التحكم في الجودة والدقة تحديًا آخر.

v. **Variability**: In addition to the increasing velocities and varieties of data, data flows can be highly inconsistent with periodic peaks. Big Data is also variable because of the multitude of data dimensions resulting from multiple disparate data types and sources e.g. Data in bulk could create confusion whereas less amount of data could convey half or Incomplete Information. Variability of data can be challenging to manage.

التباين: بالإضافة إلى السرعات المتزايدة وأنواع البيانات، يمكن أن تكون تدفقات البيانات غير متسقة. كما أن البيانات الضخمة متغيرة بسبب تعدد أبعاد البيانات الناتجة عن أنواع ومصادر بيانات متعددة ومتباينة، على سبيل المثال، يمكن أن تؤدي البيانات المجمعة إلى حدوث ارتباك في حين أن كمية أقل من البيانات يمكن أن تنقل نصف المعلومات أو معلومات غير كاملة. يمكن أن يكون إدارة تباين البيانات أمرًا صعبًا.

vi. **Value**: The bulk Data having no Value is of no good to the company, unless you turn it into something useful. Data in itself is of no use or importance but it needs to be converted into something valuable to extract Information. The ability to transform the bulk data into business and make this enormous data of use to business to monetize. It is a challenge to connect and correlate relationships, hierarchies and multiple data linkages of the big data.

القيمة: إن البيانات الضخمة التي لا قيمة لها لا تقدم أي فائدة للشركة، ما لم تحولها إلى شيء مفيد. فالبيانات في حد ذاتها لا فائدة منها ولا أهمية لها، ولكن يجب تحويلها إلى شيء ذي قيمة لاستخراج المعلومات. والقدرة على تحويل البيانات الضخمة إلى عمل تجاري وجعل هذه البيانات الضخمة مفيدة للشركات لتحقيق الربح منها. إن ربط وترابط العلاقات والتسلسلات الهرمية وارتباطات البيانات المتعددة للبيانات الضخمة يمثل تحديًا.

1.4 FOUNDATION OF DATA SCIENCE

Data science involves using methods to analyse massive amounts of data and extract the knowledge it contains. Data Science is an interdisciplinary field focused on extracting knowledge from datasets which are large in size, applying the knowledge from data to solve problems in a wide range of application domains. Mathematics, statistics, computer science, and domain knowledge are the foundations of Data Science.

The main components of Data Science are given below:

1. **Statistics**:
   To analyze the large amount numerical data and to find the meaningful insights from it, knowledge of statistics is required
2. **Domain Expertise**:
   Data is available and applicable in various domains, therefore domain expertise or specialized knowledge of a specific area is required to get best results.
3. **Data engineering:**
   Data engineering is a part of data science, which involves acquiring, storing, retrieving, and transforming the data. Data engineering also includes metadata (data about data) to the data.

4. **Visualization:**
   Data visualization is meant by representing data in a visual context so that people can easily understand the significance of data. Data visualization makes it easy to access the huge amount of data in visuals.

5. **Advanced computing:**
   Advanced computing involves designing, writing, debugging, and maintaining the source code of computer programs. Machine Learning and Deep learning techniques are required for modelling and to make predictions about unforeseen/future data.

1.5 TOOLS FOR DATA SCIENCE

Some tools required for data science are as follows:

- **Data Analysis tools**: R, Python, Statistics, SAS, Jupyter, R Studio, MATLAB, Excel, RapidMiner.
- **Data Warehousing**: ETL, SQL, Hadoop, Informatica/Talend, AWS Redshift
- **Data Visualization tools**: R, Jupyter, Tableau, Cognos.
- **Machine learning tools**: Spark, Mahout, Azure ML studio.

1.6 APPLICATIONS OF DATA SCIENCE

Data Science is used by most organizations for predictive analysis, price optimization, and customer satisfaction. Some areas where data science is being used are Health Care, Finance, Security, Airline Routing, Manufacturing, Speech Recognition, Advertisement, Security, Fraud detection, Banking, Internet of Things etc.

Some use cases are given below:

- Genomic Data provides deeper understanding of Genetic issues and reactions to particular drugs and diseases. توفر البيانات الجينومية فهمًا أعمق للقضايا الجينية وردود الفعل تجاه أدوية وأمراض معينة.
- Logistics companies like DHL,Fedex have discovered the best time and routes to ship cost effectively.
  اكتشفت شركات الخدمات اللوجستية مثل DHL و Fedex أفضل الأوقات والطرق للشحن بتكلفة فعالة.
- Predict employee attrition and understand the variables that influence employee turnover.
  التنبؤ بتسرب الموظفين وفهم المتغيرات التي تؤثر على دوران الموظفين.
- Airline companies can now easily predict flight delays and notify the passengers before time.
  تستطيع شركات الطيران الآن التنبؤ بسهولة بتأخيرات الرحلات وإخطار الركاب قبل الموعد المحدد.
- Banks can make better decisions by predict risk analysis, fraud detection and better customer management

يمكن للبنوك اتخاذ قرارات أفضل من خلال التنبؤ بتحليل المخاطر واكتشاف الاحتيال وإدارة العملاء بشكل أفضل

1.7 DATA SCIENCE PROCESS

The structured approach to data science helps in maximizing the chances of success in a data science project at the lowest cost. **The data science process typically consists of the following steps:**

I. **Business Understanding**: The main purpose here is making sure all the stakeholders understand the **what**, **how**, and **why** of the project. It involves two main steps namely defining research goals and preparing a project charter.

   i) **Defining Research Goals**: It is very essential to understand the business goals to be able to define the research goal that states the purpose of assignment in a clear and focused manner. The data by for the same can be gathered by repeatedly asking questions and enquiring about business expectations, identifying the research goals so that everybody knows what to do and can agree on the best course of action. The outcome should be a clear research goal, a good understanding of the context, well-defined deliverables, and a plan of action with a timetable. This information is then best placed in a project charter.

   ii) **Creating Project Charter**: A project charter has the information gathered while setting the research goal. The project charter must contain a clear research goal, the project mission and context, method for analysis, information about the resources required for project completion, proof that the project is achievable, or proof of concepts, deliverables and a measure of success and also a timeline for the project. This information is useful to make an estimation of the project costs and the data and people required for the project to become a success.

II. **Data Acquisition** or **Data Collection** or **Data Retrieval:** This phase involves data gathering from various sources like webservers, logs, databases, API's and online repositories. It involves acquiring data from all the identified internal & external sources. One should start with data collection from internal sources. The data may be available in many forms ranging from simple text to database records. The data may be structured as well as unstructured. Data may be acquired from sources outsise the organization also. Some sources may be available free of cost or some may be paid. Data collection is a tiresome and time-consuming task.

III. **Data preparation**: Data collection is an error-prone process; in this phase you enhance the quality of the data and prepare it for use in subsequent steps. This phase consists of three subphases: **data cleansing, data integration data transformation**.

i) **Data Cleansing** involves cleansing of data collected in the previous phase. It involves removing false values from a data source, removing inconsistencies across data sources, checking misspelled attributes, missing and duplicate values and typing errors, fixing capital letter mismatches as most programming languages are case sensitive (e.g India and india), removing redundant white spaces, sanity checks (check for impossible values like six-digit phone number etc) and outliers (an outlier is an observation that seems to be distant from other observations). It should be ensured that most errors are corrected in this phase to make the data usable and get better results.

ii) **Data Integration** enriches data sources by combining information from multiple data sources. Data comes from several sources and in this sub step the focus is on integrating these different sources. The data can be combined in the following ways:

- **Joining Tables**: To combine information about some data in one table with the information available in another table. e.g a table may contain information about purchase of a particular product and the other table may contain information about people who have purchased that product. This can be done using join command in SQL.
- **Appending** or **Stacking**: To add observations from one table to another table e.g. a table may contain the information about the purchase of a particular product in the year 2000 and the other contains the similar data in the year 2001, then appending means to add the records of 2001 to the table containing the records of 2000. This can be done using the union function in SQL.
- **View**: View in SQL can be used to virtually combine two tables. This saves on the space requirement
- **Aggregate Functions**: Aggregate functions may be used as per requirement for combining data.

iii) **Data transformation** ensures that the data is in a suitable format to be used in the project model. Sometimes, the number of variables may have to be reduced, It involves modification of data so that it takes a suitable shape. Tools like talend and informatica can be used for transformation.

IV. **Exploratory Data Analysis (EDA)** or **Data exploration**: Data exploration is concerned with building a deeper understanding of the data. It involves the understanding of how variables interact with each other, the distribution of the data, and whether there are outliers. It involves defining and refining the selection of feature variables that will be used in model development. This is the most important step as it involves understanding of the data which will further be used for modelling. Various techniques like visualization, tabulation, clustering, and other modelling techniques can be used for exploratory analysis.

V. **Data Modelling** or **Model building**: Model building is an iterative process. In this phase, domain knowledge, and insights about the data are used for modelling. It involves identifying the data model that best fits the business requirement. For this, various machine learning techniques like KNN, Naïve's, decision tree etc may be applied on data. The model is trained on the data set and the testing of the selected 7 model is done. Data modelling can be done using Python, R, SAS. Most models consist of the following main steps:

i) **Selection of a modelling technique and variables** to enter in the model: After the exploratory analysis, the variables to be used are known, so in this phase, the variables can be used to build the model. A model that suits the project requirement has to be chosen.

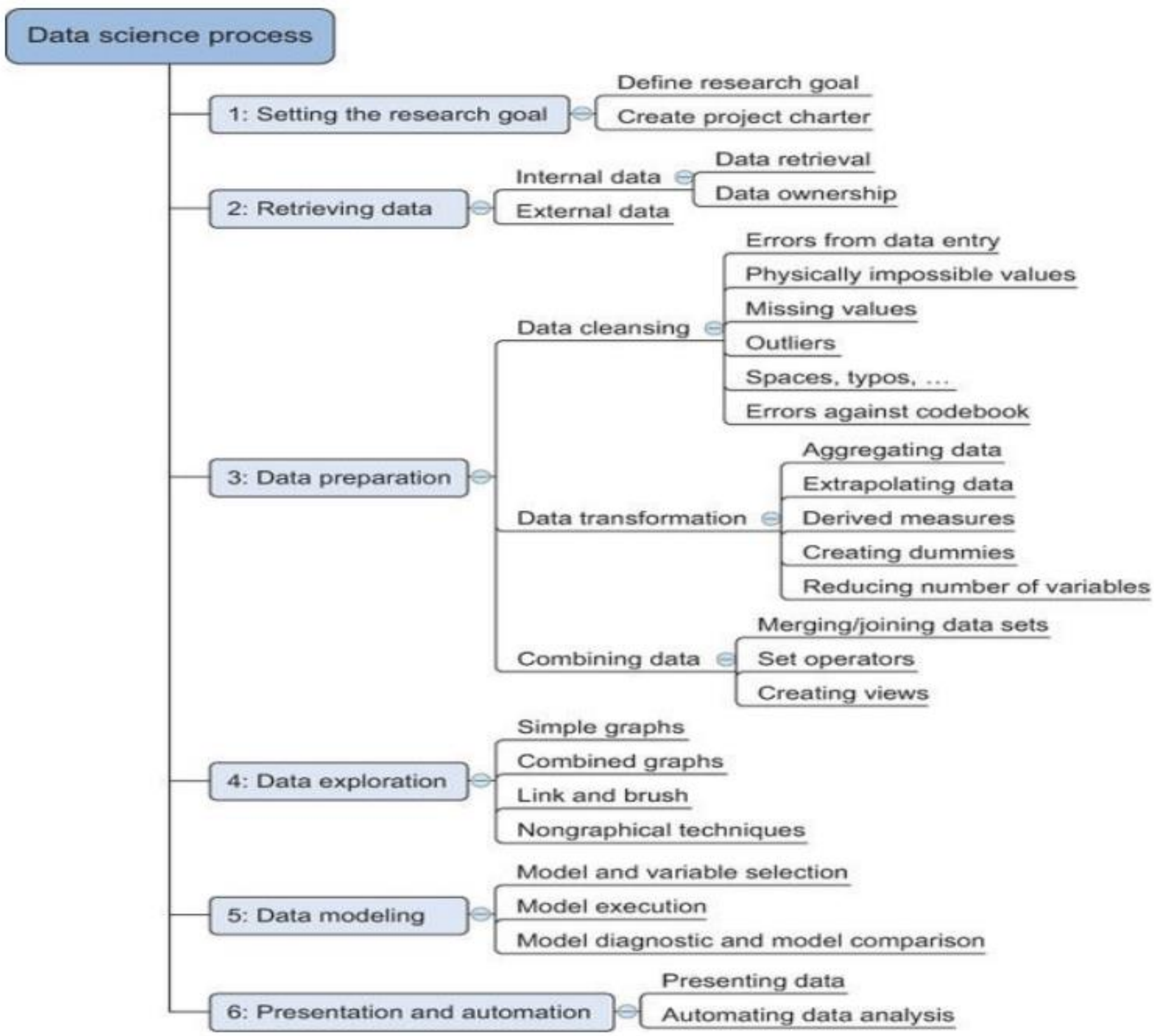
ii) **Execution of the model**: The chosen model has to be implemented by coding. Python is most used language for coding as it has many inbuilt libraries.

iii) **Diagnosis** and **model comparison**: In this step, the best performing model or the model with lowest errors is chosen from among the multiple models that are built

VI. **Presentation and automation**: In this stage, the results are presented. These results can take many forms, ranging from presentations to research reports.

VII. **Deployment & Maintenance**: Before the final deployment in the production environment, testing is done in preproduction environment. After deployment, the reports are used for real time analysis.

The Data Science **life cycle** is an Iterative process, there is often a need to step back and rework certain findings. If the step 1(business understanding) is performed dedicatedly, rework can be prevented. The figure 1 below describes the Data Science model.



1.8 MESSY DATA فوضوية

Data that is not in usable form or it is impossible to obtain clearly interpretable information from it is messy data.

البيانات التي ليست في شكل قابل للاستخدام أو من المستحيل الحصول على معلومات واضحة قابلة للتفسير منها هي بيانات فوضوية.

Data science and its algorithms are **clean** and **precise**, but the data on which they operate come from the real world, are inherently messy i.e. the data which requires some preparation before you can use them effectively and it is not easy to find clean data. The quality of insights you derive from data depends on the validity of that data, so some preparation is required. Some examples of messy data are missing data, unstructured data, multiple variables in one column, variables stored in wrong places, observations split incorrectly or left together against normalization rules, switched columns and rows, extra spaces etc.

إن علم البيانات وخوارزمياته نظيف ودقيق، ولكن البيانات التي تعمل عليها تأتي من العالم الحقيقي، وهي فوضوية بطبيعتها، أي البيانات التي تتطلب بعض التحضير قبل أن تتمكن من استخدامها بشكل فعال وليس من السهل العثور على بيانات نظيفة. تعتمد جودة الأفكار التي تستمدها من البيانات على صحة تلك البيانات، لذا يلزم بعض التحضير. بعض الأمثلة على البيانات الفوضوية هي البيانات المفقودة، والبيانات غير المنظمة، والمتغيرات المتعددة في عمود واحد، والمتغيرات المخزنة في أماكن خاطئة، والملاحظات المقسمة بشكل غير صحيح أو المتروكة معًا ضد قواعد التطبيع، والأعمدة والصفوف المبدلة، والمسافات الإضافية وما إلى ذلك.

1.9 ANOMALIES AND ARTEFACTS IN DATASETS

Anomaly is a deviation in data from the expected value for a metric at a given point in time. Anomaly detection is any process that finds the outliers (items that do not belong to the dataset) of a dataset. The term anomaly is also referred to as outlier. Outliers are the data objects that stand out among other objects in the data set and do not conform to the normal behavior in a data set. There are three kinds of anomalies namely: point anomaly, contextual anomaly, and collective anomalies.

- **Point Anomaly**: If a single instance in a given dataset is different from others with respect to its attributes, it is called a point anomaly i.e. when a single instance of data is anomalous, it deviates largely from the rest of the set e.g. detecting credit card fraud based on “amount spent.”
- Contextual anomaly: If the data is anomalous in some context, it is called contextual anomaly. This type of anomaly is common in time-series data. In the absence of a context, all the data points look normal. E.g. if the context of the temperature is recorder in December and a high temperature reading is seen in December month, which is an abnormal phenomenon.
- Collective anomalies can be formed due to a combination of many instances i.e. a set of data instances collectively helps in detecting anomalies. For example, sequence data in network log or an attempt to copy data form a remote machine to a local host unexpectedly, an anomaly that would be flagged as a potential cyber-attack.

Anomaly detection refers to the problem of finding patterns in data that do not conform to expected behavior. It is a technique for finding an unusual point or pattern in a given set. These nonconforming patterns are often referred to as anomalies, outliers, discordant observations, exceptions, aberrations, surprises, peculiarities, or contaminants in different application domains. Anomaly detection is commonly used for:

- Data cleaning
- Intrusion detection
- Fraud detection
- Systems health monitoring
- Event detection in sensor networks
- Ecosystem disturbances

Artifact It is a data flaw caused by equipment, techniques or conditions. Common sources of data flaws include hardware or software errors, conditions such as electromagnetic interference and flawed designs such as an algorithm prone to miscalculations. Some common data artifacts are:

- Digital Artifacts: Flaws in digital media, documents and data records caused by data processing errors e.g a distorted camera recording.
- Visual Artifacts: Flaws in visualizations such as user interfaces.
- Compression Artifacts: Flaws in data due to lossy compression.
- Statistical Artifacts: Flaw such as a bias in statistical data.
- Sonic Artifacts: Unwanted sound in a recording.

1.10 CLEANING DATA

Data cleaning is a key part of data science. Clean data increases overall productivity and allows for the highest quality information in decision-making. Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset. If data is messy, the outcomes and algorithms are unreliable. It can lead to poor business strategy and decision-making. The data can have many irrelevant and missing parts. To handle this part, data cleaning is done. It involves handling of missing data, noisy data etc. During cleansing, missing values may either be filled or removed depending on the data. Data cleaning is discussed later.

1.11 QUESTIONS

1. What are the foundations of data science?
2. Discuss the areas where data science is applicable?
3. Discuss the various challenges due to growth in data.
4. Explain briefly the data Science Process.
5. Define Messy data.

Python

Define a function:

```
def my_func(x, y):
    if x > y:
        return x
```

Define a function that returns a tuple:

```
def my_func(x, y):
    return (x-1, y+2)
(a, b) = my_func(1, 2)
```

```
a = 0; b = 4
```

len([‘c’, ‘m’, ‘s’, ‘c’, 3, 2, 0])   #7

range: returns an iterable object

list(range(10)

enumerate: returns iterable tuple (index, element) of a list

enumerate([“311”, “320”, “330”])

[(0, “311”), (1, “320”), (2, “330”)]

map: apply a function to a sequence or iterable

```
arr = [1, 2, 3, 4, 5]
map(lambda x: x**2, arr)
```

```
[1, 4, 9, 16, 25]
```

filter: returns a list of elements for which a predicate is true

```
arr = [1, 2, 3, 4, 5, 6, 7]
filter(lambda x: x % 2 == 0, arr)
```

```
[2, 4, 6]
```

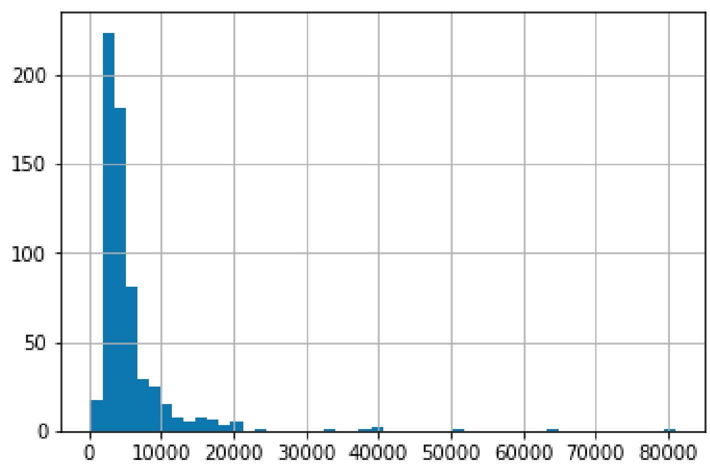
import pandas as pd
import numpy as np
import matplotlib as plt
df=pd.read_csv(“ …”)
df.head(5)
df.describe()

In [6]:
```
df['Property_Area'].value_counts()
```

Out[6]:
```
Semiurban    233
Urban        202
Rural        179
Name: Property_Area, dtype: int64
```

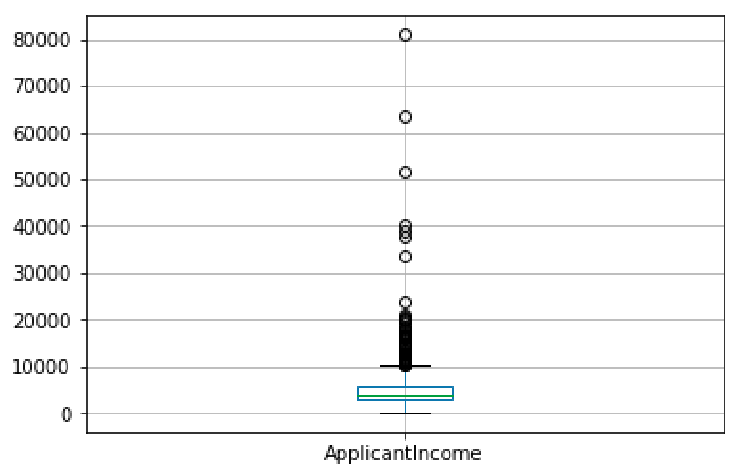
In [7]:
```
df['ApplicantIncome'].hist(bins=50)
```

Out[7]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x7f5029ab7400>
```



In [8]:
```
df.boxplot(column='ApplicantIncome')
```

Out[8]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x7f5029a7f080>
```

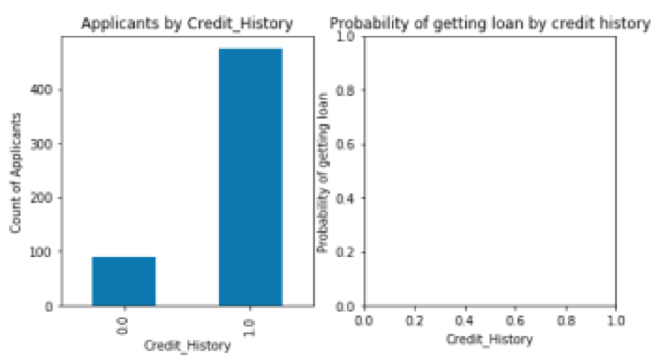


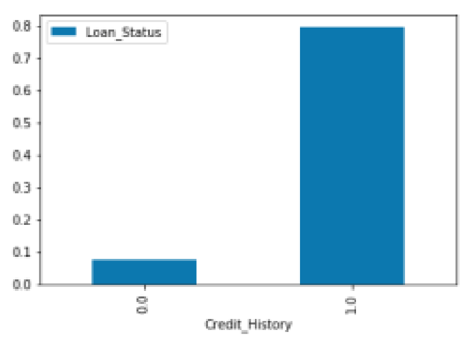
In [14]:
```
temp1 = df['Credit_History'].value_counts(ascending=True)
temp2 = df.pivot_table(values='Loan_Status',index=['Credit_History'],aggfunc=lambda x: x.map({'Y':1,'N':0}).mean())
print ('Frequency Table for Credit History:')
print (temp1)

print ('\nProbility of getting loan for each Credit History class:')
print (temp2)
```

```
Frequency Table for Credit History:
0.0     89
1.0    475
Name: Credit_History, dtype: int64

Probility of getting loan for each Credit History class:
                Loan_Status
Credit_History
0.0                0.078652
1.0                0.795789
```

In [15]:
```
import matplotlib.pyplot as plt
fig = plt.figure(figsize=(8,4))
ax1 = fig.add_subplot(121)
ax1.set_xlabel('Credit_History')
ax1.set_ylabel('Count of Applicants')
ax1.set_title("Applicants by Credit_History")
temp1.plot(kind='bar')

ax2 = fig.add_subplot(122)
temp2.plot(kind = 'bar')
ax2.set_xlabel('Credit_History')
ax2.set_ylabel('Probability of getting loan')
ax2.set_title("Probability of getting loan by credit history")
```

Out[15]: `Text(0.5,1,'Probability of getting loan by credit history')`





Course Structure:

Lectures 1-2: introduction & Python Data Science

Lecture 3: data collection & management

Best Practices, data wrangling, exploratory analysis, ethics, debugging, visualization, …

Lectures 4-5: Statistical modeling & ML

Statistical learning, regression, classification, cross validation, model evaluation, hypothesis testing, ….

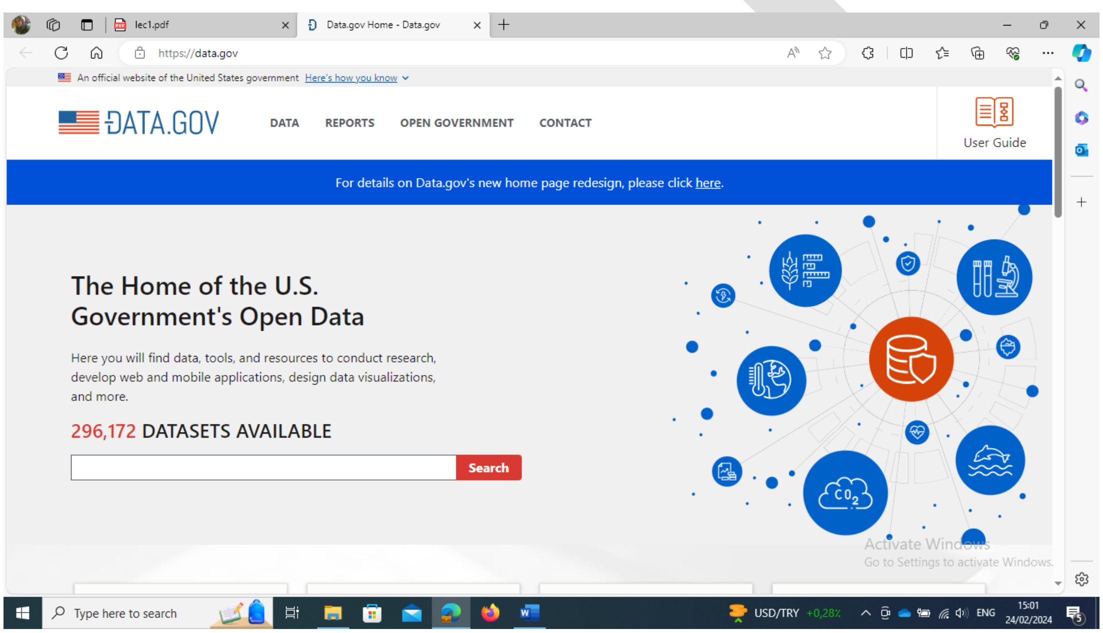
MidTerm Exam

Lectures 6-8: Advanced Topics

Dimensionality reduction, distributed Learning, big data, distributed computation

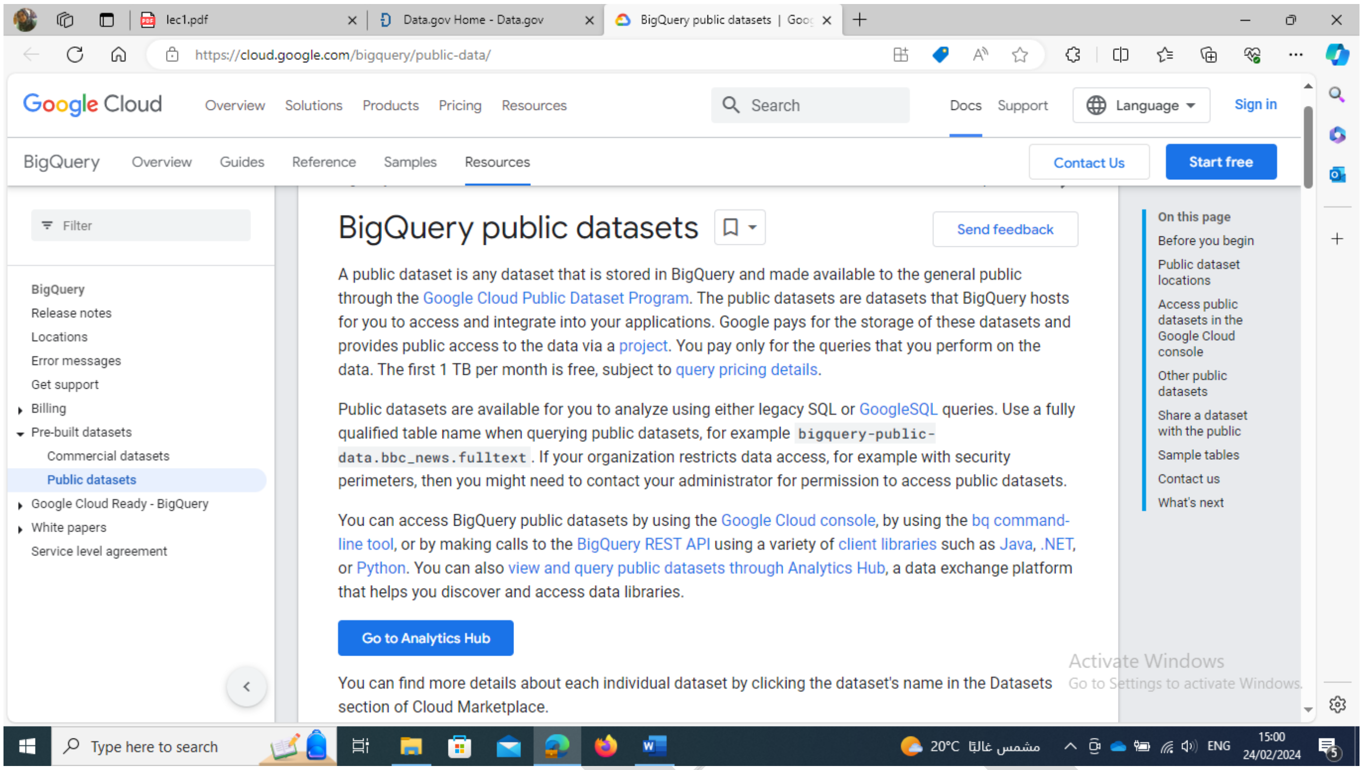
Datasets resources:

1- www.data.gov/
agriculture, climate, education, energy, finance, health, …



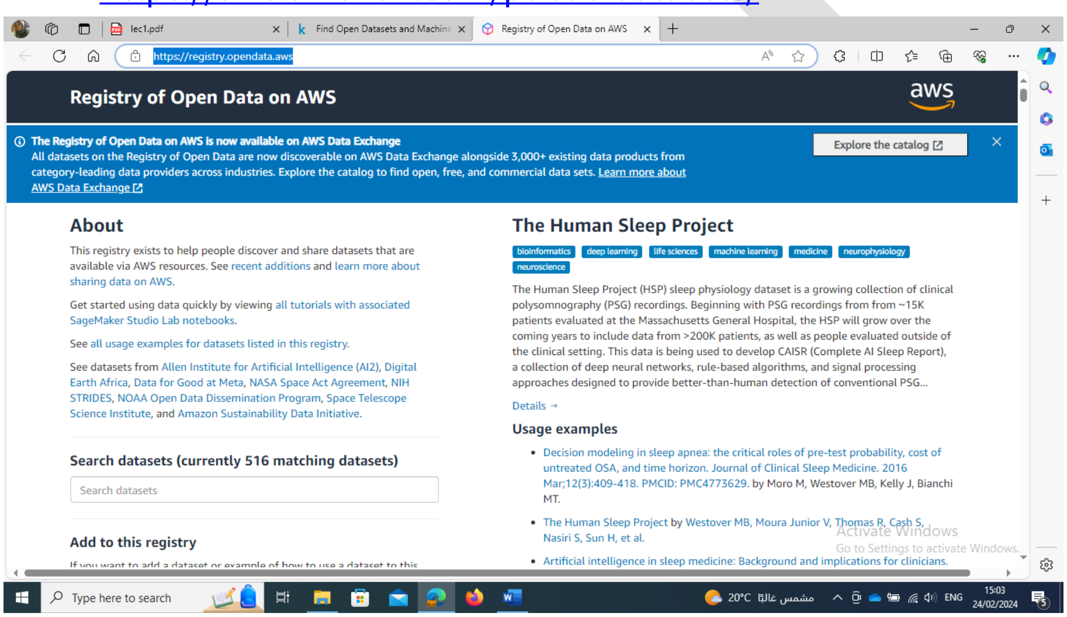
2- https://cloud.google.com/bigquery/public-data/

BigQuery (google cloud) public datasets.



3- www.kaggle.com/datasets

4- https://aws.amazon.com/public-datasets/



References

1- Joel Grus, "Data Science from Scratch First Principals with Python", 2nd edition, O'REILLY, USA, 2019.

2- Laura Lagual and Santi Segui,"**Introduction to Data Science** A Python Approach to Concepts, Techniques and Applications", Springer 2017